

Thermodynamic State Vector Interface & Nonequilibrium Energy Equations in Planetary Simulation Engines

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Abstract

As planetary-scale ecosystem simulations scale across complex biogeochemical cycles (carbon, nitrogen, phosphorus, and water), maintaining strict mathematical compliance with the First and Second Laws of Thermodynamics is paramount. This paper details the architectural implementation of the *Thermodynamic State Vector Interface* established in Sprint 16 of the *Web of Life* simulation architecture. We formalize internal entropy generation ($\dot{S}_{\text{gen}}$), exergy destruction rate quantification via the Gouy-Stodola theorem ($\dot{I} = T_0 \dot{S}_{\text{gen}}$), and boundary heat and mass flux arrays. Furthermore, we introduce functional monad state transitions that enforce mass conservation, non-negative entropy generation, and solar-only external energy input constraints at every simulation time step.

Official Repository: <https://github.com/pascalranoroarijaona/WebOfLife>

1 Introduction and Objectives

The *Web of Life* simulation engine models complex biogeochemical cycles across four interacting planetary spheres: Atmosphere, Hydrosphere, Lithosphere, and Biosphere. To ensure long-term stability and physical realism, unphysical energy creation or spontaneous entropy reduction must be programmatically prevented.

Sprint 16 establishes rigorous software and mathematical contracts for:

1. Strict type definitions for internal entropy generation ($\dot{S}_{\text{gen}}$).
2. Exergy destruction rate quantification ($\dot{I} = T_0 \dot{S}_{\text{gen}}$).
3. Comprehensive boundary heat and mass flux vector arrays.
4. Functional monad state transitions enforcing mass conservation and solar-only energy inputs.

2 Thermodynamic Foundations

2.1 First Law: Conservation of Energy

The total energy change within any sub-system or planetary boundary Ω is governed by:

$$\frac{dE}{dt} = \sum_k \dot{Q}_k - \dot{W} + \sum_i \dot{m}_i h_i \quad (1)$$

Within our architecture, **all external energy input must originate solely from solar radiation** ($\dot{Q}_{\text{solar}}$), while internal work ($\dot{W}$) and boundary heat losses obey strict open-system accounting.

2.2 Second Law: Entropy Generation & Exergy Destruction

The Second Law mandates that entropy within an interacting control volume cannot spontaneously decrease without boundary compensation:

$$\frac{dS}{dt} = \sum_k \frac{\dot{Q}_k}{T_k} + \sum_i \dot{m}_i s_i + \dot{S}_{\text{gen}} \quad (2)$$

Where the internal entropy generation rate $\dot{S}_{\text{gen}} \geq 0$ under all nonequilibrium conditions. The **Exergy Destruction Rate** ($\dot{I}$), representing thermodynamic irreversibility, is defined via the Gouy-Stodola theorem relative to an ambient reference temperature $T_0 = 288.15 \text{ K}$:

$$\dot{I} = T_0 \dot{S}_{\text{gen}} \quad (3)$$

3 State Transition Equations

For a discrete simulation time step Δt , state transitions applied via the thermodynamic monad compute delta stocks as follows:

$$\Delta U = \left(\sum_k \dot{Q}_k - \dot{W} + \sum_i \dot{m}_i h_i \right) \Delta t \quad (4)$$

$$\Delta S = \left(\sum_k \frac{\dot{Q}_k}{T_k} + \sum_i \dot{m}_i s_i + \dot{S}_{\text{gen}} \right) \Delta t \quad (5)$$

$$\dot{I} = T_0 \dot{S}_{\text{gen}} \quad (6)$$

4 Software Implementation

The TypeScript interface contracts in `src/thermodynamics/types.ts` enforce these physical laws at compile-time and runtime. The `ThermodynamicStateMonad` intercepts all state mutations, throwing an architecture fault if $\dot{S}_{\text{gen}} < 0$ or if exergy destruction deviates from $T_0 \dot{S}_{\text{gen}}$.

5 Conclusion

Sprint 16 successfully bridges rigorous nonequilibrium thermodynamics with planetary-scale multi-agent simulation. By embedding physical laws directly into the type system and monad transition pipelines, the *Web of Life* engine provides a robust platform for ecological and thermodynamic research.

Source code and supplementary materials are available at <https://github.com/pascalranoroarijaona/WebOfLife>.